

Lab 8: Fast Decay with RC Circuits

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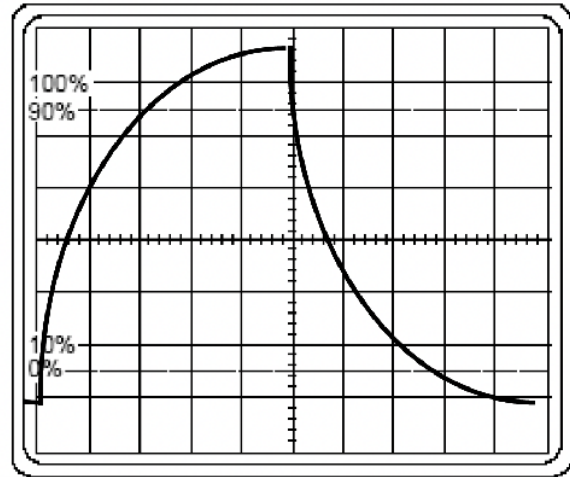
Professor Szwerc

Pre-lab Questions

1. You have constructed an RC circuit and you see the trace to the right on your oscilloscope screen. You have set the vertical scale to 1 V per square and the horizontal to 20 ms per square. What is the time constant of the RC circuit?

The time constant $\tau = RC$ is equivalent to the “1/e life” of the voltage across the capacitor. That is, at what time t after discharging does the voltage across the capacitor drop to $1/e \approx 36.8\%$ of its initial value?

The range of the voltage is from the line at -10% to the line at 110% . $1/e \approx 36.8\%$ between -10% and 110% is approximately 34.2% , which intersects with the curve about 1 division past the y-axis.



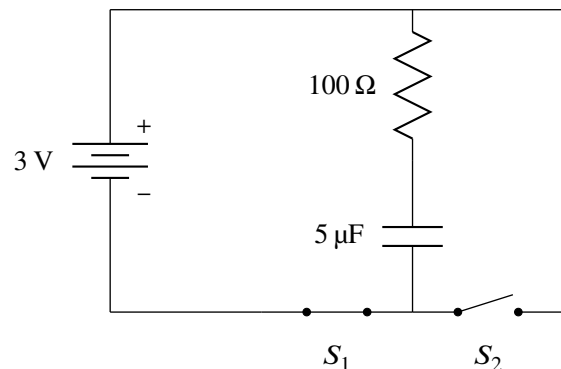
This corresponds to a time constant of approximately:

$$\tau \approx 1 \text{ division} \cdot 20 \text{ ms/division} = 20 \text{ ms.}$$

2. The RC circuit is constructed as shown on the right.
 - b. You then change the resistance to 200Ω . How does the time constant change?

The time constant is proportional to the resistance: $\tau \propto R$. Thus, doubling the resistance will double the time constant:

$$\tau' = 2\tau.$$



- b. You then change the capacitance to $20 \mu\text{F}$. How does the time constant change?

The time constant is proportional to the capacitance: $\tau \propto C$. Thus, quadrupling the capacitance will quadruple the time constant:

$$\tau' = 4\tau.$$

- c. You then change to a 9 V battery. How does the time constant change?

The time constant is independent of voltage, so the time constant will not change:

$$\tau' = \tau.$$

Procedure

1. An RC circuit was constructed with a $5600\ \Omega$ resistor and a $0.1\ \mu\text{F}$ capacitor.

Observed half-life:

$$t_{1/2, \text{observed}} = 1 \frac{\text{ms}}{\text{division}} \cdot 0.4 \text{ divisions} = 0.4 \text{ ms.}$$

Calculated half-life:

$$t_{1/2, \text{calculated}} = RC \ln 2 = 5600\ \Omega \cdot 0.1\ \mu\text{F} \cdot \ln 2 = 0.388 \text{ ms.}$$

Percent error: 3.05%

2. Switching resistance and capacitance:

- a. The old resistor is swapped with a $10\ 000\ \Omega$ resistor, and the same $0.1\ \mu\text{F}$ capacitor is used.

Observed half-life:

$$t_{1/2, \text{observed}} = 1 \frac{\text{ms}}{\text{division}} \cdot 0.6 \text{ divisions} = 0.6 \text{ ms.}$$

Calculated half-life:

$$t_{1/2, \text{calculated}} = RC \ln 2 = 10\ 000\ \Omega \cdot 0.1\ \mu\text{F} \cdot \ln 2 = 0.693 \text{ ms.}$$

Percent error: 13.44%

- b. The old capacitor is swapped with a $0.05\ \mu\text{F}$ capacitor¹, and the $10\ 000\ \Omega$ resistor from part (a) is used. We chose a capacitance that was half of the previous capacitance to “reverse” the effect of (approximately) doubling the resistance in part (a).

Observed half-life:

$$t_{1/2, \text{observed}} = 1 \frac{\text{ms}}{\text{division}} \cdot 0.36 \text{ divisions} = 0.36 \text{ ms.}$$

Calculated half-life:

$$t_{1/2, \text{calculated}} = RC \ln 2 = 10\ 000\ \Omega \cdot 0.05\ \mu\text{F} \cdot \ln 2 = 0.347 \text{ ms.}$$

Percent error: 3.87%

3. The resistor is replaced with a potentiometer with maximum resistance of $10\ 000\ \Omega$. At an arbitrary position of the potentiometer, the observed half-life is:

$$t_{1/2} \approx 0.5 \frac{\text{ms}}{\text{division}} \cdot 0.2 \text{ divisions} = 0.1 \text{ ms.}$$

From this observed half-life, we can calculate the resistance of the potentiometer at this position:

$$t_{1/2} = RC \ln 2 \rightarrow R = \frac{t_{1/2}}{C \ln 2} = \frac{0.1 \text{ ms}}{0.05\ \mu\text{F} \cdot \ln 2} = 2885\ \Omega.$$

Measured Resistance with DMM: $3360\ \Omega$ (15.20% difference)

¹Since there was no $0.05\ \mu\text{F}$ capacitor, we connected two $0.1\ \mu\text{F}$ capacitors in series to get an equivalent capacitance of $0.05\ \mu\text{F}$.

Conclusion

Examine the shapes of the exponential voltage signals across the resistor. Give a qualitative explanation for this shape in a manner similar to that presented above (in the METHOD section) for the RC circuit. Do these two voltages in fact add up to a square wave as demanded by Kirchoff's rule? Explain your answer!